Old Exam Questions
Unsupervised Machine Learning

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How this file is organized

This file is separate from the course summary and is meant as a clean practice set. Exact duplicates were removed. When essentially the same question appeared in multiple old exams, that is marked with *Asked multiple times.* inside the question box. The wording was lightly normalized where the scans were messy, but the mathematical content and answer choices were kept intact.

Contents

Probability, Information Theory, and Estimation

Question 1: Cumulants for independent uniform variables

Assume you have two independent random variables X and Y which are uniformly distributed on $[-1, 1]$. As in the lecture, let $\kappa_n(X)$ denote the n-th order cumulant of a random variable X. Compute the following quantities and answer the last classification question. Use the hint

$$\kappa_4(X) = \mathbb{E}(X^4) - 3\big(\mathbb{E}(X^2)\big)^2$$

for centered X.

- $\kappa_4(X) =$ ________________
- $\kappa_4(2X) =$ ________________
- $\kappa_4(X + Y) =$ ________________
- X is sub-Gaussian / super-Gaussian: ______________________

Question 2: Cumulants for symmetric Bernoulli variables

Assume you have two independent, discrete random variables X and Y, both following independent symmetric Bernoulli distributions taking only the values $+1$ and -1, each with probability $1/2$. As in the lecture, let $\kappa_n(X)$ denote the n-th order cumulant of a random variable X. Compute the quantities below. Use the hint

$$\kappa_3(X) = \mathbb{E}(X^3), \qquad \kappa_4(X) = \mathbb{E}(X^4) - 3\big(\mathbb{E}(X^2)\big)^2$$

for centered X.

- $\kappa_3(X) =$ ________________
- $\kappa_4(X) =$ ________________
- $\kappa_4(3X)/\kappa_4(X) =$ ________________
- $\kappa_4(X + Y)/\kappa_4(X) =$ ________________

Question 3: Eigenvectors, covariance matrix, and a scalar expression

You are given a dataset with n samples and three features. The covariance matrix C has eigenvalues $(\lambda_1, \lambda_2, \lambda_3) = (5, 3, 1)$ and associated normalized eigenvectors (v_1, v_2, v_3) with $\|v_i\| = 1$. Compute

$$A = (2v_1^\top + v_2^\top + 3v_3^\top)\, C\, (3v_2 + v_3).$$

Enter the final scalar value:

$$A = \_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_$$

Question 4: KL divergence for continuous uniform distributions

Asked multiple times.

Assume Y and Z are uniformly distributed on $[0, \theta_Y]$ and $[0, \theta_Z]$, respectively:

$$p_Y(x) = \begin{cases} \frac{1}{\theta_Y}, & x \in [0, \theta_Y], \\ 0, & \text{else}, \end{cases} \qquad p_Z(x) = \begin{cases} \frac{1}{\theta_Z}, & x \in [0, \theta_Z], \\ 0, & \text{else}. \end{cases}$$

For $\theta_Y = e^5$ and $\theta_Z = e^{10}$, compute

$$D_{\mathrm{KL}}(p_Y \| p_Z) = \int_{-\infty}^{\infty} p_Y(x) \log\left(\frac{p_Y(x)}{p_Z(x)}\right) \mathrm{d}x.$$

Use the natural logarithm and round to two decimal places.

$$D_{\mathrm{KL}}(p_Y \| p_Z) = \underline{\hspace{3cm}}$$

Question 5: Outer product matrix

Given two vectors

$$x = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \qquad y = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix},$$

compute the outer product matrix

$$A = xy^\top.$$

Enter the integer values of the third row of A:

$$A_{31} = \underline{\hspace{2cm}} \qquad A_{32} = \underline{\hspace{2cm}} \qquad A_{33} = \underline{\hspace{2cm}}.$$

Question 6: Marginal probabilities, entropies, and mutual information — variant 1

Asked multiple times.

Assume you are given two binary random variables X and Y taking only the values 0 and 1 where their joint probability distribution is specified by

	$Y = 0$	$Y = 1$
$X = 0$	0.25	0.125
$X = 1$	0.25	0.375

This means, for example, that $P(X = 0, Y = 0) = 0.25$. Compute the quantities below using the natural logarithm. Round the final values to two decimal places.

- $P(X = 1) =$ ____________
- $P(Y = 1) =$ ____________
- $H(X) =$ ____________
- $H(Y) =$ ____________
- $I(X; Y) =$ ____________

Question 7: Marginal probabilities, entropies, and mutual information — variant 2

Asked multiple times.

Assume you are given two binary random variables X and Y taking only the values 0 and 1 where their joint probability distribution is specified by

	$Y = 0$	$Y = 1$
$X = 0$	$\frac{1}{3}$	$\frac{1}{3}$
$X = 1$	0	$\frac{1}{3}$

This means, for example, that $P(X = 0, Y = 0) = \frac{1}{3}$. Compute the quantities below using the natural logarithm. Round the final values to two decimal places.

- $P(Y = 0) =$ ____________
- $H(X) =$ ____________
- $H(Y) =$ ____________
- $I(X; Y) =$ ____________

Question 8: Marginal probabilities, entropies, and mutual information — variant 3

Asked multiple times.

Assume you are given two binary random variables X and Y taking only the values 0 and 1 where their joint probability distribution is specified by

	$Y = 0$	$Y = 1$
$X = 0$	$\frac{3}{10}$	$\frac{4}{10}$
$X = 1$	$\frac{2}{10}$	$\frac{1}{10}$

Compute the following quantities using the natural logarithm. Round the final values to two decimal places.

- $P(X = 0) =$ ____________
- $P(Y = 0) =$ ____________
- $H(X) =$ ____________
- $H(Y) =$ ____________
- $I(X; Y) =$ ____________

Question 9: Parameter Estimation and Maximum Likelihood

Asked multiple times.

Which statements about Parameter Estimation and Maximum Likelihood are true? Multiple answers are possible.

- ☐ If the data are distributed i.i.d., the likelihood can be written as a product of individual probabilities over the data samples.
- ☐ The likelihood and the log-likelihood are maximized for the same parameter value, and at that parameter value $\hat{w}$, their values coincide: $\mathcal{L}(\hat{w}) = \ln \mathcal{L}(\hat{w})$.
- ☐ Parameter estimation and the maximum likelihood approach rest on the assumption that the model class is not known and estimated from the data.
- ☐ The maximum likelihood estimator is asymptotically unbiased and efficient, but not asymptotically consistent.
- ☐ The mean squared error can be written as the sum of the variance of the estimator and the squared bias of the estimator.
- ☐ An estimator $\hat{w}$ is unbiased if its expected value equals the true parameter: $\mathbb{E}(\hat{w}) = w$.
- ☐ The likelihood $\mathcal{L}(\{x\}; w)$ is the probability that the model with parameter w produces the data $\{x\}$.
- ☐ The mean squared error is defined as $\mathrm{mse}(\hat{w}) = (\hat{w} - w)^\top(\hat{w} - w)$.

Question 10: Dropdown selection task on estimation theory

Asked multiple times.

Fill in the correct choices.

For any parameter w, an estimator $\hat{w}$ is *consistent / efficient / unbiased* if $\mathbb{E}(\hat{w}) = w$. The *expected value / bias / mean squared error / variance* of $\hat{w}$ can be decomposed as a sum of its *expected value / bias / mean squared error / variance* and its squared *expected value / bias / mean squared error / variance.*

Let $X_1, \ldots, X_n$ be random variables drawn independently from a distribution $p(\cdot; w)$. Consider

$$\frac{1}{n+1} \sum_{i=1}^{n} X_i$$

as an estimator for the mean of the distribution $p(\cdot; w)$. This estimator is *neither unbiased nor consistent / unbiased but not consistent / biased but consistent / both unbiased and consistent.*

Now consider X_1 as an estimator for the mean of the distribution $p(\cdot; w)$. This estimator is *neither unbiased nor consistent / unbiased but not consistent / biased but consistent / both unbiased and consistent.*

Let $\{x_1, \ldots, x_n\}$ be a sample drawn independently from the distribution $p(\cdot; w)$. The function

$$\prod_{i=1}^{n} p(x_i; w)$$

is called the *probability / loss / density / likelihood* function. Since $p(\cdot; w)$ is usually a *probability / loss / density / likelihood* function, the probability measure of the set $\{x_1, \ldots, x_n\}$ is in fact zero.

The Cramér–Rao lower bound relates the *expected value / bias / mean squared error / variance* of the estimator $\hat{w}$ to the inverse of its Fisher information. An estimator that reaches the Cramér–Rao lower bound is said to be *consistent / efficient / unbiased.*

Question 11: Unsupervised Machine Learning

Which statements about Unsupervised Machine Learning are true? Multiple answers are possible.

- ☐ Underfitting means that a model is adapted to special training characteristics such as noisy measurements or outliers.
- ☐ Projection methods project data into a higher-dimensional space while keeping their neighborhoods.
- ☐ The model selection procedure is called “training” or “learning”.
- ☐ Examples of unsupervised methods are: PCA, ICA, Factor Analysis, k-means clustering, and mixture models.
- ☐ In parametric models, every parameter (vector) realization represents a model.
- ☐ k-nearest neighbors is a nonparametric model.
- ☐ Generative models do not aim at modeling the data creation process.
- ☐ Unsupervised Machine Learning algorithms learn patterns from unlabeled data.

Question 12: Expectation Maximization (EM) and Maximum Entropy — version A

Which statements about Expectation Maximization (EM) and Maximum Entropy are correct? Multiple answers are possible.

- ☐ The EM algorithm can be applied to hidden Markov models and mixture models.
- ☐ In the M-step, the model parameters are updated.
- ☐ The E-step increases the upper bound on the log-likelihood.
- ☐ In EM, one deals with problems that have hidden variables as well as model parameters.
- ☐ The entropy of a probability distribution is always larger equal 0 and smaller equal 1.
- ☐ Hidden state models in general have analytical solutions for the likelihood.
- ☐ In the E-step, the hidden variables parameters are updated.
- ☐ The Maximum Entropy approach requires minimal prior assumptions to assign a-priori probabilities.

Question 13: Expectation Maximization (EM) and Maximum Entropy — version B

Asked multiple times.

Which statements about Expectation Maximization (EM) and Maximum Entropy are correct? Multiple answers are possible.

- ☐ In the M-step, the hidden variables are updated.
- ☐ Maximum Entropy requires minimal prior assumptions to assign a-priori probabilities.
- ☐ The E-step increases the upper bound on the log-likelihood.
- ☐ In the E-step, the model parameters are updated.
- ☐ The EM algorithm can be applied to hidden Markov models and mixture models.
- ☐ The entropy of a probability distribution is always larger equal 0 and smaller equal 1.
- ☐ The E-step increases the lower bound on the log-likelihood.
- ☐ In EM, one deals with problems that have hidden variables as well as model parameters.

PCA, Kernel PCA, and ICA

Question 14: Principal Component Analysis — statements, version A

Which statements about Principal Component Analysis (PCA) are correct? Multiple answers are possible.

- ☐ The first principal component corresponds to the largest eigenvalue in the eigendecomposition of the sample covariance matrix.
- ☐ The singular values from the singular value decomposition of the data matrix X are equal to the eigenvalues of the eigendecomposition of $X^\top X$.
- ☐ The PCA solution is unique, i.e. there is only one solution.
- ☐ Eigenvectors can be computed iteratively with Oja's rule.

- ☐ The sample variance of the k-th projection is equal to the k-th eigenvalue of the sample covariance matrix.
- ☐ The matrix of eigenvectors U is an orthogonal matrix.
- ☐ The sample covariance matrix is symmetric and positive semidefinite.
- ☐ PCA transforms data into a new coordinate system such that the data has the largest mean value along the first coordinate, the second largest mean value along the second coordinate, and so on.

Question 15: Principal Component Analysis — statements, version B

We keep the notation of the lecture slides: X is the data matrix and U is the matrix of eigenvectors of $X^\top X$. Which statements about PCA are true? Multiple answers are possible.

- ☐ PCA transforms data into a new coordinate system such that the data has the largest variance along the first coordinate, the second largest variance along the second coordinate, and so on.
- ☐ The first principal component corresponds to the largest eigenvalue in the eigendecomposition of the sample covariance matrix.
- ☐ If there are n data samples, each of which is d-dimensional, the matrix U has dimension $n \times d$.
- ☐ The sample covariance matrix is orthogonal and positive semidefinite.
- ☐ The singular values from the singular value decomposition of the data matrix X are equal to the square root of the eigenvalues of the eigendecomposition of $X^\top X$.
- ☐ The eigenvalues obtained from the eigendecomposition of $X^\top X$ are all positive and sum up to 1.
- ☐ PCA is commonly used to visualize data by downprojecting it to a smaller number of dimensions, keeping the most relevant information.
- ☐ The matrix U is a symmetric matrix.

Question 16: Picture-based interpretation question: PCA, ICA, and EM

The original exam contained three picture-based subquestions. State which interpretation is more likely in each case.

1. PCA and ICA are applied to a 2D cloud of points, leading to outcomes A and B. Which picture is more likely the result of PCA and which one the result of ICA?

   PCA: ________________ ICA: ________________

2. Two plots show a dataset in a coordinate system spanned by principal components. Which is more likely: picture 1 is shown in PC2 vs. PC1 and picture 2 in PC3 vs. PC1, or vice versa?

   Answer: ______________________________________

3. The left figure shows an original dataset, while figures a and b show results of running EM for a mixture of two Gaussians. Which is more likely: figure a is after 5 iterations and figure b after 20 iterations, or the other way around?

   Answer: ______________________________________

Question 17: PCA computation on a rank-1 matrix — projected second sample

Asked multiple times.

You are given four data samples, each of dimension 3. The data matrix reads

$$X = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \\ 4 & 4 & 4 \end{pmatrix}.$$

The eigenvectors of $X^\top X$ are

$$u_1 = \frac{1}{\sqrt{3}}(1,1,1)^\top, \qquad u_2 = \frac{1}{\sqrt{2}}(-1,0,1)^\top, \qquad u_3 = \frac{1}{\sqrt{2}}(-1,1,0)^\top.$$

They are already normalized and already correctly ordered corresponding to eigenvalues in descending order. Compute the PCA projection matrix Y and enter the components of the projected second feature vector y_2. Round to two decimal places.

$y_2^{(1)} =$ __________ $\qquad y_2^{(2)} =$ __________ $\qquad y_2^{(3)} =$ __________

Question 18: PCA computation on a centered matrix — projected third sample

Asked multiple times.

You are given four data samples, each of dimension 3. The data matrix reads

$$X = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -3 & 0 & 3 \\ -4 & 0 & 4 \end{pmatrix}.$$

The eigenvectors of the covariance matrix are

$$u_1 = \frac{1}{\sqrt{2}}(-1,0,1)^\top, \qquad u_2 = \frac{1}{\sqrt{2}}(1,0,1)^\top, \qquad u_3 = (0,1,0)^\top.$$

They are already normalized and already correctly ordered corresponding to eigenvalues in descending order. Compute the PCA projection matrix Y and enter the components of the projected third sample y_3. Round to two decimal places.

$y_3^{(1)} =$ __________ $\qquad y_3^{(2)} =$ __________ $\qquad y_3^{(3)} =$ __________

Question 19: PCA computation with a repeated sample — projected third sample

You are given four data samples, each of dimension 3, collected in the data matrix

$$X = \begin{pmatrix} -4 & 0 & -4 \\ -2 & 0 & 2 \\ 3 & 0 & 3 \\ 3 & 0 & 3 \end{pmatrix}.$$

Further assume that the eigenvectors of the covariance matrix of this data are

$$u_1 = \frac{1}{\sqrt{2}}(1,0,1)^\top, \qquad u_2 = \frac{1}{\sqrt{2}}(-1,0,1)^\top, \qquad u_3 = (0,1,0)^\top.$$

They are already normalized and already correctly ordered corresponding to eigenvalues in descending order. Compute the PCA projection matrix Y and enter the components of the projected third sample y_3. Round to two decimal places.

$y_3^{(1)} =$ ____________ $y_3^{(2)} =$ ____________ $y_3^{(3)} =$ ____________

Question 20: Kernel PCA

Which statements about Kernel PCA are correct? Use the notation from the lecture: data points $x_1, \ldots, x_n \in \mathcal{X} = \mathbb{R}^m$, kernel $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$, feature map $\phi : \mathcal{X} \to \mathcal{H}$, feature covariance matrix

$$C = \frac{1}{n} \sum_{i=1}^{n} \phi(x_i)\phi(x_i)^\top,$$

and Gram matrix K with entries $K_{ij} = \phi(x_i)^\top \phi(x_j)$. Multiple answers are possible.

- ☐ A disadvantage of Kernel PCA is that ϕ always needs to be computed explicitly, which is computationally costly.
- ☐ In general, any nonnegative and symmetric function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ defines a kernel.
- ☐ To compute the projection of a data point x onto the eigenvectors of K, you need to center x and K, as well as to normalize the eigenvectors of K.
- ☐ The main task of Kernel PCA is to solve the eigenvalue problem of C in feature space. This can be achieved by solving the eigenvalue problem of K in a Euclidean space.
- ☐ In general, ϕ is a nonlinear map that maps into a higher-dimensional feature space $\mathcal{H}$.
- ☐ Linear PCA can be seen as a special case of Kernel PCA by using $k(x, y) = x^\top y$ as the associated kernel function.
- ☐ For Kernel PCA, it is essential to center K before computing its eigendecomposition, since the eigenvalues of K can in general be negative, which leads to inconclusive results.
- ☐ K and C always have the same dimensionality and the same rank.

Question 21: Independent Component Analysis — version A

Which statements about Independent Component Analysis (ICA) are correct? Multiple answers are possible.

- ☐ Maximizing the negentropy maximizes the mutual information.
- ☐ In general, ICA components are neither ranked nor orthogonal.
- ☐ Non-Gaussianity can be measured by the second cumulant.
- ☐ Criteria for independence are mutual information between components and non-Gaussianity of components.
- ☐ The ICA solution is unique.
- ☐ For ICA, one needs at least as many sources as observations.
- ☐ Infomax maximizes the mutual information between components.
- ☐ FastICA relies on whitening and rotating the data.

Question 22: Independent Component Analysis — version B

Which statements about Independent Component Analysis (ICA) are correct? Multiple answers are possible.

- ☐ Independence can be quantified by the mutual information between components or the non-Gaussianity of components.
- ☐ Fast ICA uses whitening and rotating the data.
- ☐ Maximizing the negentropy minimizes the mutual information.
- ☐ For non-Gaussian sources, the ICA solution is unique up to permutation and scaling.
- ☐ Non-Gaussianity can be measured by the second and third cumulant.
- ☐ For ICA, the number of sources must be smaller than, or equal to, the number of observations.
- ☐ ICA components are ranked and orthogonal.
- ☐ Infomax maximizes the mutual information between components.

Question 23: Independent Component Analysis — version C

Asked multiple times.

Which statements about Independent Component Analysis (ICA) are correct? Multiple answers are possible.

- ☐ For ICA, one needs more sources than observations.
- ☐ Maximizing the negentropy maximizes the mutual information.
- ☐ Criteria for independence are mutual information between components and non-Gaussianity of components.
- ☐ The ICA solution is unique.
- ☐ Fast ICA relies on whitening and rotating the data.
- ☐ Infomax maximizes the mutual information between components.
- ☐ Non-Gaussianity can be measured by the second cumulant.
- ☐ In general, ICA components are neither ranked nor orthogonal.

Factor Analysis

Question 24: Factor Analysis: low-rank covariance part

In factor analysis, part of the covariance function is formed by

$$C_U = UU^\top = \sum_{i=1}^{\ell} u_i u_i^\top,$$

where $U \in \mathbb{R}^{m\times\ell}$, the vectors u_i are the columns of U, and $\ell < m$. Which constraints does this impose on C_U? Multiple answers are possible.

- ☐ The maximum rank of the matrix is ℓ.

- ☐ The number of zero eigenvalues of C_U is greater or equal to $m - \ell$.
- ☐ Any covariance matrix in m dimensions can be written in the form above.
- ☐ C_U is singular.

Question 25: Factor Analysis: dimensions of the objects

We keep the notation from the lecture. Here n is the number of data samples, m is the dimension of each sample, ℓ is the number of factors, y is the factor vector, U is the factor loading matrix, Ψ is the diagonal noise covariance matrix, and ε is the noise vector. Which statements are correct? Multiple answers are possible.

- ☐ Ψ has shape $m \times m$.
- ☐ Ψ has shape $\ell \times \ell$.
- ☐ Ψ has shape $n \times \ell$.
- ☐ U has shape $m \times \ell$.
- ☐ U has shape $m \times n$.
- ☐ U has shape $\ell \times m$.
- ☐ ε has dimension m.
- ☐ ε has dimension ℓ.
- ☐ ε has dimension n.
- ☐ y has dimension m.
- ☐ y has dimension ℓ.
- ☐ y has dimension n.

Question 26: Factor Analysis: compute the first row of the covariance matrix

You are given the factor loading matrix

$$U = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$$

and the diagonal noise covariance matrix

$$\Psi = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix}.$$

Compute the first row of the covariance matrix

$$C = UU^\top + \Psi.$$

Enter integer values:

$C_{11} =$ __________ $C_{12} =$ __________ $C_{13} =$ __________

Question 27: Factor Analysis: conceptual statements

Which statements about Factor Analysis (FA) are correct? Multiple answers are possible.

- ☐ The factors are unique only up to orthogonal transformations (rotations).
- ☐ Correlations between observations can only be explained by factors since the noise covariance matrix Ψ is diagonal.
- ☐ FA describes the variability of observations in terms of latent variables and noise.
- ☐ The factor loading matrix U describes the independent part of the covariance of the observations.
- ☐ FA decomposes the covariance matrix of the data as $C = UU^\top + \Psi$.
- ☐ FA assumes Gaussian noise and non-Gaussian latent variables.
- ☐ Factors explain the variance of the observations, while the correlation is explained by noise.

Clustering, Mixtures, and Projection Methods

Question 28: k-means: one iteration, dataset A

Assume you want to do k-means on the following 2-dimensional dataset:

$$X = \{(1,0), (0,1), (-1,0), (0,-1), (2,0), (2,3), (5,0)\}.$$

The initial cluster centers are $\mu = (0,1)$ and $\nu = (2,2)$. Compute the coordinates of the new cluster centers μ^{new} and ν^{new} after one iteration of k-means.

$$\mu^{\text{new}} = (\underline{\hspace{2cm}}, \underline{\hspace{2cm}}), \qquad \nu^{\text{new}} = (\underline{\hspace{2cm}}, \underline{\hspace{2cm}})$$

Question 29: k-means and affinity propagation

Which statements about k-means and affinity propagation are correct? Multiple answers are possible.

- ☐ The k-means objective is to minimize the within-cluster sum of squared deviations.
- ☐ In affinity propagation: the larger the preference, the smaller is the number of cluster centers.
- ☐ In affinity propagation: the larger the damping factor, the faster the convergence.
- ☐ In affinity propagation, cluster centers are data points.
- ☐ In affinity propagation, the similarity between two points is typically chosen as their negative squared distance.
- ☐ k-means always converges to the global optimum.
- ☐ The parameter k in k-means is automatically chosen by the algorithm to best fulfill the optimization objective.
- ☐ k-means is sensitive to initialization of the cluster centers.

Question 30: k-means: one iteration, dataset B

Assume you want to do k-means on the following 2-dimensional dataset:

$$X = \{(1,0),(2,-1),(1,-2),(0,-1),(2,0),(2,1),(3,0),(3,1)\}.$$

The initial cluster centers are $\mu = (0,-2)$ and $\nu = (4,1)$. Compute the coordinates of the new cluster centers μ^{new} and ν^{new} after one iteration of k-means.

$$\mu^{\text{new}} = (\underline{\hspace{2cm}}, \underline{\hspace{2cm}}), \qquad \nu^{\text{new}} = (\underline{\hspace{2cm}}, \underline{\hspace{2cm}})$$

Question 31: k-means: one iteration, dataset C

Assume you want to do k-means on the following 2-dimensional dataset:

$$X = \{(1,-1),(0,0),(-1,-1),(0,-2),(2,0),(2,3),(5,0)\}.$$

The initial cluster centers are $\mu = (0,0)$ and $\nu = (2,1)$. Compute the coordinates of the new cluster centers μ^{new} and ν^{new} after one iteration of k-means.

$$\mu^{\text{new}} = (\underline{\hspace{2cm}}, \underline{\hspace{2cm}}), \qquad \nu^{\text{new}} = (\underline{\hspace{2cm}}, \underline{\hspace{2cm}})$$

Question 32: k-means: one iteration, dataset D

Assume you want to do k-means on the following 2-dimensional dataset:

$$X = \{(0,1),(1,2),(2,1),(1,0),(0,2),(-1,2),(0,3),(-1,3)\}.$$

The initial cluster centers are $\mu = (2,0)$ and $\nu = (-1,4)$. Compute the coordinates of the new cluster centers μ^{new} and ν^{new} after one iteration of k-means. A sketch of the data points helps: the assignment step can largely be read off graphically. Give the values with one decimal place.

$$\mu^{\text{new}} = (\underline{\hspace{2cm}}, \underline{\hspace{2cm}}), \qquad \nu^{\text{new}} = (\underline{\hspace{2cm}}, \underline{\hspace{2cm}})$$

Question 33: t-SNE

Which statements about t-distributed Stochastic Neighborhood Embedding are correct? Multiple answers are possible.

- ☐ It models the neighborhood in both observation and embedding space by a Gaussian distribution.
- ☐ It minimizes the Kullback–Leibler divergence between distributions in observation space and embedding space.
- ☐ It models the neighborhood of a data point in observation space proportional to a Student's t-distribution.
- ☐ It is usually trained by gradient descent.

Question 34: Scaling and Projection Methods — version A

Which statements about Scaling and Projection Methods are correct? Multiple answers are possible.

- ☐ Isomap uses a geodesic distance matrix, whose eigenvectors are the coordinates of the projected space.
- ☐ The objective in t-SNE is minimization of the KL divergence between the similarities of the high-dimensional inputs and the low-dimensional mapped outputs.
- ☐ In locally linear embedding (LLE), each data point is described as a linear combination of its neighbors.
- ☐ The objective of scaling is to project data into a higher-dimensional space, e.g. to allow better model selection.
- ☐ In self-organizing maps the objective is to minimize an energy (error) function.
- ☐ Multidimensional scaling (MDS) keeps the distances between data points as well as possible.
- ☐ LLE in general performs linear mappings to realize neighborhood-preserving embeddings.
- ☐ In t-SNE, the similarity between data points i and j is given by their joint probability.

Question 35: Projection and scaling methods — version B

Which statements about projection and scaling methods are correct? Multiple answers are possible.

- ☐ Non-negative Matrix Factorization (NMF) performs a multiplicative decomposition of a positive semidefinite matrix into two positive semidefinite matrices.
- ☐ Locally Linear Embedding (LLE) is a linear embedding method that aims to preserve the neighborhood of each data point.
- ☐ Multidimensional Scaling (MDS) aims to find a low-dimensional representation of the data that preserves pairwise Euclidean distances.
- ☐ Stochastic Neighbor Embedding (SNE) defines the similarity of two data points as a conditional probability while t-SNE defines it as a joint probability.
- ☐ Isomap measures distances between data points as the length of the shortest path along a neighborhood graph.

Question 36: Scaling and projection methods — version C

Tick the correct statements about scaling and projection methods. We collect the original data points in a matrix X and the mapped points in a matrix Y. Multiple answers are possible.

- ☐ The t-distribution has heavier tails than the Gaussian distribution; thus in t-SNE the joint distributions q_{ij} between y_i and y_j are computed using a t-distribution in order to alleviate the crowding problem.
- ☐ Isomap tries to approximate the geodesic distance between two points on a manifold by constructing a neighborhood graph.
- ☐ In SNE and t-SNE, the perplexity term is computed using the entropy of the similarities q_{ij} of the mapped data points y_i and y_j.
- ☐ In SNE and t-SNE, the objective is the mean squared distance between x_i and y_i.
- ☐ Locally linear embedding and multidimensional scaling are probabilistic methods that

rely on minimizing the KL divergence between the distributions of X and Y.

- ☐ In SNE and t-SNE, the computation of the similarity p_{ij} between x_i and x_j relies on assuming a Gaussian distribution centered at x_i.

Question 37: Scaling and Projection Methods — version D

Asked multiple times.

Which statements about Scaling and Projection Methods are correct? Multiple answers are possible.

- ☐ Multidimensional scaling (MDS) keeps the distances between data points as well as possible.
- ☐ The objective of scaling is to project data into a higher-dimensional space, e.g. to allow better model selection.
- ☐ Isomap uses a geodesic distance matrix, whose eigenvectors are the coordinates of the projected space.
- ☐ In t-SNE, the similarity between data points i and j is given by their joint probability.
- ☐ In self-organizing maps the objective is to minimize an energy (error) function.
- ☐ The objective in t-SNE is minimization of the KL divergence between the similarities of the high-dimensional inputs and the low-dimensional mapped outputs.
- ☐ In locally linear embedding (LLE), each data point is described as a linear combination of its neighbors.
- ☐ LLE in general performs linear mappings to realize neighborhood-preserving embeddings.

Question 38: Scaling and Projection Methods — version E

Decide which statements about Scaling and Projection Methods are correct. Multiple answers are possible.

- ☐ The objective of scaling and projection methods is to project data into a lower-dimensional space, e.g. to allow better model selection.
- ☐ In self-organizing maps the objective is to minimize an energy (error) function.
- ☐ Projection pursuit performs projections such that the extracted signal is as Gaussian as possible.
- ☐ Generative Topographic Mapping (GTM) is a non-linear latent variable model which maps latent variables to observations.
- ☐ Isomap uses a geodesic distance matrix, whose largest eigenvectors are the coordinates of the projected space.
- ☐ The objective in t-SNE is minimization of the KL divergence between the similarities of the (high-dimensional) inputs and the (low-dimensional) mapped outputs.
- ☐ In locally linear embedding (LLE), each data point is described as a linear combination of its neighbors.
- ☐ Multidimensional Scaling (MDS) projects data to a higher-dimensional space in order to increase the distance between data points.

Question 39: Non-negative Matrix Factorization (NMF)

As in the lecture, let

$$X = YU^\top, \qquad Y \in \mathbb{R}^{n\times\ell}, \qquad U \in \mathbb{R}^{m\times\ell},$$

with non-negativity constraints. Which statements about NMF are correct? Multiple answers are possible.

- ☐ NMF objectives are the KL divergence and the Euclidean distance.
- ☐ NMF is always exactly solvable.
- ☐ $X \in \mathbb{R}^{n\times m}$.
- ☐ The non-negativity constraints make the representation of the observations purely additive.
- ☐ The outer product representation reads $X = \sum_{k=1}^{\ell} y_k u_k^\top$.
- ☐ NMF factorizes any positive or negative definite matrix into two positive definite matrices.

Question 40: Mixturetown bread question

In Mixturetown there are two bakers, Albert (A) and Bibi (B). Their loaf weights are modeled as Gaussian distributions

$$f(x; \mu_A, \sigma_A), \qquad f(x; \mu_B, \sigma_B),$$

with

$$\mu_A = 1 \text{ kg}, \quad \sigma_A = 0.1 \text{ kg}, \qquad \mu_B = 1.5 \text{ kg}, \quad \sigma_B = 0.2 \text{ kg}.$$

Seventy-five percent of all loaves are made by Bibi. You observe a loaf of weight 1.1 kg. Determine the probability that Bibi made this loaf. Use the given density values

$$f(1.1; 1, 0.1) = 2.41971, \qquad f(1.1; 1.5, 0.2) = 0.26995.$$

Round to two decimal places.

$$P(B \mid x = 1.1) = \underline{\qquad\qquad}$$

Question 41: Mixture model posterior probability

Asked multiple times.

Assume the following case:

- The prior probability of component j is 0.1.
- The probability for observing a data sample x given the total parameter set θ is 0.2.
- The likelihood for observing a data sample x, given component j and corresponding parameter vector, is 0.5.

Compute the posterior probability of component j. Round to two decimal places.

$$P(j \mid x) = \underline{\qquad\qquad}$$

Question 42: Mixture model: infer the likelihood from posterior, prior, and evidence

Asked multiple times.

Assume the following case:

- The prior probability of component j is 0.1.
- The posterior probability of component j is 0.3.
- The probability for observing a data sample x given the total parameter set θ is 0.2.

Compute the likelihood for observing a data sample x given component j and the corresponding parameter vector. Round to two decimal places.

$$P(x \mid j) = \underline{\hspace{3cm}}$$

Question 43: Mixture of Gaussians: posterior of the first Gaussian

The data generation process in a mixture of Gaussians model is as follows: first choose one Gaussian according to its weight in the mixture, then draw a sample from that Gaussian. You have fitted a mixture of two Gaussians to 1D data:

$$p(x) = \alpha_1 f(x; \mu_1, \sigma_1) + \alpha_2 f(x; \mu_2, \sigma_2),$$

with

$$\alpha_1 = 0.3, \quad \alpha_2 = 0.7, \quad \mu_1 = 0, \quad \sigma_1 = 1, \quad \mu_2 = 3, \quad \sigma_2 = 1.$$

Relevant densities are

$$f(1; 0, 1) = 0.242, \qquad f(1; 3, 1) = 0.054.$$

Calculate the probability that the datapoint $x = 1$ has been produced by the first Gaussian. Round to two decimal places.

$$P(G_1 \mid x = 1) = \underline{\hspace{3cm}}$$

Question 44: KL divergence for continuous uniform distributions — variant 2

Assume Y and Z are uniformly distributed on $[0, \theta_Y]$ and $[0, \theta_Z]$, respectively. For $\theta_Y = e^4$ and $\theta_Z = e^{10}$, compute

$$D_{\mathrm{KL}}(p_Y \| p_Z) = \int_{-\infty}^{\infty} p_Y(x) \log\left(\frac{p_Y(x)}{p_Z(x)}\right) \mathrm{d}x.$$

Use the natural logarithm and round to two decimal places.

$$D_{\mathrm{KL}}(p_Y \| p_Z) = \underline{\hspace{3cm}}$$

Question 45: KL divergence for two discrete distributions

Asked multiple times.

You are given two probability distributions p and q:

$$p(x = 0) = 0.8, \quad p(x = 1) = 0.2, \qquad q(x = 0) = 0.5, \quad q(x = 1) = 0.5.$$

Compute the KL divergence

$$D_{\mathrm{KL}}(p \| q).$$

Use the natural logarithm and round to two decimal places.

$$D_{\mathrm{KL}}(p\|q) = \underline{\hspace{3cm}}$$

Question 46: Clustering — version A

Which statements about Clustering are correct? Multiple answers are possible.

- ☐ In affinity propagation, the cluster centers are always data points.
- ☐ Clustering (or Cluster Analysis) is both a regression and a classification technique.
- ☐ In both k-means and fuzzy k-means, the membership distribution is always given by a binary random variable.
- ☐ In mixture models, an observation is assigned to the component with the largest prior distribution.
- ☐ Top-down clustering relies on building the minimal spanning tree and then in each step removes the largest of the remaining edges.
- ☐ In a mixture of Gaussians, the maximum likelihood and the maximum posterior always have the same value.
- ☐ The agglomerative hierarchical clustering algorithm repeatedly fuses the two maximally distant clusters.
- ☐ Similarity-based clustering does not require objects to be represented via feature vectors.

Question 47: Clustering — version B

Which statements about Clustering are correct? Multiple answers are possible.

- ☐ k-means is faster than MoG and not sensitive to initialization.
- ☐ Both k-means and MoG assign an observation to a single cluster only.
- ☐ The mixing coefficients in a Mixture of Gaussians model are represented by the posteriors.
- ☐ The maximum a posteriori estimation is a way to avoid near-zero variances and singular solutions in the maximum likelihood estimation of mixture models.
- ☐ Given an observation x, a component j, and a parameter vector θ, the posterior $p(j \mid x, \theta)$ describes how likely it is that x was produced by j.
- ☐ The EM algorithm for MoG tries to maximize the likelihood with respect to the cluster centers, the covariance matrices, and the mixing coefficients.
- ☐ The k-means algorithm is a simplification of the Mixture of Gaussians model.
- ☐ Clustering is an unsupervised technique that aims at finding separate regions of high data density called clusters.

Question 48: Hierarchical clustering: single linkage vs. complete linkage

You are given 8 datapoints in $\mathbb{R}^2$:

$$\{(0,1),(1,1),(2,1),(8,1),(0,-1),(1,-1),(2,-2),(8,-1)\}.$$

Agglomerative hierarchical clustering is run until only two clusters remain, once with single linkage and once with complete linkage. Two resulting clusterings are shown in the exam figure. Check the correct statements.

- ☐ Clustering 1 has been obtained by single linkage.
- ☐ Clustering 2 has been obtained by single linkage.
- ☐ Clustering 2 has been obtained by complete linkage.
- ☐ Clustering 1 has been obtained by complete linkage.
- ☐ One of the two clusterings could theoretically have been the result of either linkage function.

Question 49: Hierarchical clustering with average linkage on letters

Asked multiple times.

You are given the set of data points

$$x = \{a, c, e, h, i, m, n, s\},$$

where the similarity between two points $s(x, x')$ is defined as the distance between the ordered letters in the alphabet, for example $s(a, b) = 1$, $s(a, c) = 2$, and so on. Perform agglomerative clustering using average linkage,

$$d_{\text{avg}}(A, B) = \frac{1}{n_A}\frac{1}{n_B}\sum_{a\in A}\sum_{b\in B} s(a, b),$$

and stop when the distance between clusters is greater than 7. Answer the following questions with integer values.

- How many clusters are formed in total? __________
- What is the maximum number of data points in any one cluster? __________
- What is the minimum distance between any two of the formed clusters? __________

Question 50: Biclustering

Which statements about Biclustering are correct? Multiple answers are possible.

- ☐ A common biclustering method is the variance maximization method.
- ☐ FABIA is a generative model.
- ☐ FABIA defines biclusters as inner products.
- ☐ In FABIA, the likelihood is analytically intractable and requires, for example, a variational EM algorithm.
- ☐ Row and column elements must belong to exactly one bicluster or to no bicluster at all.
- ☐ Biclustering simultaneously clusters the rows and columns of a matrix.

Dense Answer Key

Compact solutions

Q1. $\kappa_4(X) = -0.13$, $\kappa_4(2X) = -2.13$, $\kappa_4(X+Y) = -0.27$, sub-Gaussian.

Q2. $\kappa_3(X) = 0$, $\kappa_4(X) = -2$, $\kappa_4(3X)/\kappa_4(X) = 81$, $\kappa_4(X+Y)/\kappa_4(X) = 2$.

Q3. $A = 12$.

Q4. $D_{\mathrm{KL}}(p_Y \| p_Z) = 5.00$.

Q5. Third row $(12, 15, 18)$.

Q6. $P(X = 1) = 0.63$, $P(Y = 1) = 0.50$, $H(X) = 0.66$, $H(Y) = 0.69$, $I(X;Y) = 0.03$.

Q7. $P(Y = 0) = 0.33$, $H(X) = 0.64$, $H(Y) = 0.64$, $I(X;Y) = 0.17$.

Q8. $P(X = 0) = 0.70$, $P(Y = 0) = 0.50$, $H(X) = 0.61$, $H(Y) = 0.69$, $I(X;Y) = 0.02$.

Q9. Correct: a, e, f, g.

Q10. unbiased; mean squared error; variance; bias; biased but consistent; unbiased but not consistent; likelihood; density; variance; efficient.

Q11. Correct: d, e, f, h.

Q12. Correct: a, b, d, g, h.

Q13. Correct: b, e, g, h.

Q14. Correct: a, d, e, f, g.

Q15. Correct: a, b, e, g.

Q16. PCA corresponds to A, ICA corresponds to B; picture 1 is more likely PC2 vs. PC1 and picture 2 PC3 vs. PC1; in the EM part, a is earlier and b is later.

Q17. $y_2 = (3.46, 0.00, 0.00)$.

Q18. $y_3 = (4.24, 0.00, 0.00)$.

Q19. $y_3 = (4.24, 0.00, 0.00)$.

Q20. Correct: c, d, e, f.

Q21. Correct: b, d, h.

Q22. Correct: a, b, c, d, f.

Q23. Correct: c, e, h.

Q24. Correct: a, b, d.

Q25. Correct: a, d, g, k.

Q26. First row $[10, 11, 17]$.

Q27. Correct: a, b, c, e.

Q28. $\mu^{\text{new}} = (0, 0)$, $\nu^{\text{new}} = (3, 1)$.

Q29. Correct: a, d, e, h.

Q30. $\mu^{\text{new}} = (1, -1)$, $\nu^{\text{new}} = (2.5, 0.5)$.

Q31. $\mu^{\text{new}} = (0, -1)$, $\nu^{\text{new}} = (3, 1)$.

Q32. $\mu^{\text{new}} = (1.0, 1.0)$, $\nu^{\text{new}} = (-0.5, 2.5)$.

Q33. Correct: b, d.

Q34. Correct: a, b, c, f, h.

Q35. Correct: c, d, e.

Q36. Correct: a, b, f.

Q37. Correct: a, c, d, f, g.

Q38. Correct: a, d, e, f, g.

Q39. Correct: a, c, d, e.

Q40. $P(B \mid x = 1.1) = 0.25$.

Q41. $P(j \mid x) = 0.25$.

Q42. $P(x \mid j) = 0.60$.

Q43. $P(G_1 \mid x = 1) \approx 0.66$.

Q44. $D_{\mathrm{KL}}(p_Y \| p_Z) = 6.00$.

Q45. $D_{\mathrm{KL}}(p \| q) = 0.19$.

Q46. Correct: a, h.

Q47. Correct: d, e, f, g, h.

Q48. Correct: b, d, e.

Q49. 3 clusters; maximum cluster size 4; minimum inter-cluster distance 8.

Q50. Correct: b, d, f.